

Safety data sheet

Page: 1/10

BASF Safety data sheet
Date / Revised: 29.05.2021
Product: **Plurafac® LF 403**

Version: 5.0

(30044016/SDS_GEN_TH/EN)

Date of print 24.01.2024

1. Substance/preparation and manufacturer/supplier identification

Plurafac® LF 403

Use: Raw material for the chemical-technical industry

Manufacturer/supplier:

BASF SE
67056 Ludwigshafen
GERMANY
Operating Division Care Chemicals
Telephone: +49 621 60-57579
E-mail address: emd-ems-ehs-masterdata@basf.com

Emergency information:

International emergency number:
Telephone: +49 180 2273-112

2. Hazard identification

Classification according to UN GHS 2009

Classification of the substance and mixture:

Skin corrosion/irritation: Cat. 2

Hazardous to the aquatic environment - acute: Cat. 1

Hazardous to the aquatic environment - chronic: Cat. 3

M-factor acute: 1

Label elements and precautionary statement:

Pictogram:



Signal Word:
 Warning

Hazard Statement:

H315	Causes skin irritation.
H412	Harmful to aquatic life with long lasting effects.
H400	Very toxic to aquatic life.

Precautionary Statements (Prevention):

P280	Wear protective gloves.
P273	Avoid release to the environment.
P264	Wash contaminated body parts thoroughly after handling.

Precautionary Statements (Response):

P302 + P352	IF ON SKIN: Wash with plenty of soap and water.
P332 + P313	If skin irritation occurs: Get medical attention.
P391	Collect spillage.
P362 + P364	Take off contaminated clothing and wash it before reuse.

Precautionary Statements (Disposal):

P501	Dispose of contents and container to hazardous or special waste collection point.
------	---

Other hazards which do not result in classification:

No specific dangers known, if the regulations/notes for storage and handling are considered.

This surfactant complies with the biodegradability criteria as laid down in Regulation (EC) No.648/2004 on detergents. Data to support this assertion are held at the disposal of the competent authorities of the Member States and will be made available to them at their direct request or at the request of a detergent manufacturer.

3. Composition/information on ingredients

Chemical nature

Alcohols, C12-15-branched and linear, ethoxylated propoxylated
CAS Number: 120313-48-6

4. First-Aid Measures

General advice:

Remove contaminated clothing.

If inhaled:

Keep patient calm, remove to fresh air, seek medical attention.

On skin contact:

Wash thoroughly with soap and water. Seek medical attention.

On contact with eyes:

Wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist.

On ingestion:

Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention.

Note to physician:

Symptoms: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11., (Further) symptoms and / or effects are not known so far

Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

5. Fire-Fighting Measures

Suitable extinguishing media:
water spray, dry powder, foam

Unsuitable extinguishing media for safety reasons:
water jet

Specific hazards:

harmful vapours, carbon oxides

Evolution of fumes/fog. The substances/groups of substances mentioned can be released in case of fire.

Special protective equipment:

Wear a self-contained breathing apparatus.

Further information:

The degree of risk is governed by the burning substance and the fire conditions. Contaminated extinguishing water must be disposed of in accordance with official regulations.

6. Accidental Release Measures

Personal precautions:

Use personal protective clothing. Information regarding personal protective measures, see section 8.

Environmental precautions:

Contain contaminated water/firefighting water. Do not discharge into drains/surface waters/groundwater.

Methods for cleaning up or taking up:
For large amounts: Dike spillage. Pump off product.
For residues: Pick up with suitable absorbent material.
Dispose of absorbed material in accordance with regulations.

Additional information: High risk of slipping due to leakage/spillage of product. Forms slippery surfaces with water.

7. Handling and Storage

Handling

Protect against moisture. Shut containers immediately after taking product because product takes up the humidity of air.

Protection against fire and explosion:
Take precautionary measures against static discharges.

Storage

Suitable materials for containers: High density polyethylene (HDPE), Low density polyethylene (LDPE), Carbon steel (Iron), Stainless steel 1.4301 (V2), Stainless steel 1.4401, Aluminium, tinned carbon steel (Tinplate), Galvanized carbon steel (Zinc), glass
Further information on storage conditions: Keep container tightly closed and dry; store in a cool place.

The product is not damaged by low temperatures or by frost.
Protect from temperatures above: 70 °C
Properties of the product change irreversibly on exceeding the limit temperature.

8. Exposure controls and personal protection

Components with occupational exposure limits

| No substance specific occupational exposure limits known.

Personal protective equipment

Respiratory protection:
Respiratory protection in case of vapour/aerosol release. Particle filter with medium efficiency for solid and liquid particles (e.g. EN 143 or 149, Type P2 or FFP2)

Hand protection:

| Chemical resistant protective gloves

Suitable materials also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN ISO 374-1):
nitrile rubber (NBR) - 0.4 mm coating thickness

Supplementary note: The specifications are based on tests, literature data and information of glove manufacturers or are derived from similar substances by analogy. Due to many conditions (e.g.

(30044016/SDS_GEN_TH/EN)

Date of print 24.01.2024

temperature) it must be considered, that the practical usage of a chemical-protective glove in practice may be much shorter than the permeation time determined through testing.
 Manufacturer's directions for use should be observed because of great diversity of types.

Eye protection:
 Safety glasses with side-shields.

Body protection:
 Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

General safety and hygiene measures:
 Wearing of closed work clothing is required additionally to the stated personal protection equipment.
 No eating, drinking, smoking or tobacco use at the place of work. Handle in accordance with good industrial hygiene and safety practice.

9. Physical and Chemical Properties

Form:	liquid	
Colour:	colourless to yellowish	
Odour:	product specific	
Odour threshold:	not determined	
pH value:	approx. 7 (50 g/l, 23 °C)	(DIN EN 1262)
solidification temperature:	approx. -25 °C	(DIN 51583)
Boiling point:	> 250 °C	
Flash point:	approx. 190 °C	(ISO 2592)
Evaporation rate:	No data available.	
Flammability (solid/gas):	hardly combustible	
Lower explosion limit:	For liquids not relevant for classification and labelling., The lower explosion point may be 5 - 15 °C below the flash point.	
Upper explosion limit:	For liquids not relevant for classification and labelling.	
Ignition temperature:	> 200 °C	(DIN 51794)
Thermal decomposition:	> 300 °C	
Self ignition:	not self-igniting	
Self heating ability:	It is not a substance capable of spontaneous heating according to UN transport regulations class 4.2.	

BASF Safety data sheet
Date / Revised: 29.05.2021
Product: **Plurafac® LF 403**

Version: 5.0

(30044016/SDS_GEN_TH/EN)

Date of print 24.01.2024

Explosion hazard: not explosive
Fire promoting properties: not fire-propagating

Vapour pressure: < 0.1 hPa
(20 °C)

Density: approx. 0.94 g/cm³ (DIN 51757)
(20 °C)

Relative density: No data available.

Relative vapour density (air):
not determined

Solubility in water: insoluble

Miscibility with water:
not (e.g. <10%)

Solubility (qualitative) solvent(s): petroleum spirit, ethanol
soluble

Partitioning coefficient n-octanol/water (log Pow):
not applicable

Surface tension:
No data available.

Viscosity, dynamic: 60 mPa.s
(23 °C)

Other Information:

If necessary, information on other physical and chemical parameters is indicated in this section.

10. Stability and Reactivity

Conditions to avoid:

Avoid humidity. See SDS section 7 - Handling and storage.

Thermal decomposition: > 300 °C

Substances to avoid:

caustics, halogens, Alkalines, acids, reactive chemicals

Corrosion to metals: Corrosive effects to metal are not anticipated.

Hazardous reactions:

No hazardous reactions when stored and handled according to instructions.

Hazardous decomposition products:

No hazardous decomposition products if stored and handled as prescribed/indicated.

11. Toxicological Information

Acute toxicity

BASF Safety data sheet
Date / Revised: 29.05.2021
Product: **Plurafac® LF 403**

Version: 5.0

(30044016/SDS_GEN_TH/EN)

Date of print 24.01.2024

Assessment of acute toxicity:

Virtually nontoxic after a single ingestion. The inhalation of a highly enriched/saturated vapor-air-mixture represents an unlikely acute hazard.

Experimental/calculated data:

LD50 rat (oral): > 5,000 mg/kg

LC50 rat (by inhalation):

not determined

LD50 rat (dermal):

not determined

Irritation**Assessment of irritating effects:**

Irritating to skin. Not irritating to the eyes.

Experimental/calculated data:

Skin corrosion/irritation rabbit: Irritant. (Draize test)

Serious eye damage/irritation rabbit: non-irritant (Draize test)

Respiratory/Skin sensitization**Assessment of sensitization:**

No data available.

Germ cell mutagenicity**Assessment of mutagenicity:**

The substance was not mutagenic in bacteria.

Carcinogenicity**Assessment of carcinogenicity:**

No data available.

Reproductive toxicity**Assessment of reproduction toxicity:**

No data available.

Developmental toxicity**Assessment of teratogenicity:**

No data available.

Specific target organ toxicity (single exposure):

Remarks: No data available.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:
No data available.

Aspiration hazard

No aspiration hazard expected.

12. Ecological Information

Ecotoxicity

Assessment of aquatic toxicity:
Very toxic (acute effect) to aquatic organisms. Harmful to aquatic life with long lasting effects.
The inhibition of the degradation activity of activated sludge is not anticipated when introduced to biological treatment plants in appropriate low concentrations.

Toxicity to fish:
LC50 (96 h) > 1 - < 10 mg/l, Fish (OECD Guideline 203)

Aquatic invertebrates:
EC50 (48 h) ≤ 1 mg/l, Daphnia magna (OECD Guideline 202, part 1)

Aquatic plants:
EC50 (72 h) ≤ 1 mg/l, Scenedesmus subspicatus (OECD Guideline 201)
acute Effect

EC10 (72 h) > 0.1 - < 1 mg/l, Scenedesmus subspicatus (OECD Guideline 201)
long-term effect

Microorganisms/Effect on activated sludge:
1,000 mg/l, activated sludge (DIN EN ISO 8192)

Chronic toxicity to fish:
No data available.

Chronic toxicity to aquatic invertebrates:
No observed effect concentration (21 d), > 0.1 - < 1 mg/l, Daphnia magna

Assessment of terrestrial toxicity:
No data available concerning terrestrial toxicity.

Mobility

Assessment transport between environmental compartments:
The substance will not evaporate into the atmosphere from the water surface.
Adsorption to solid soil phase is possible.

Persistence and degradability

Assessment biodegradation and elimination (H₂O):
Readily biodegradable (according to OECD criteria).

BASF Safety data sheet
Date / Revised: 29.05.2021
Product: **Plurafac® LF 403**

Version: 5.0

(30044016/SDS_GEN_TH/EN)

Date of print 24.01.2024

Elimination information:

> 60 % CO₂ formation relative to the theoretical value (28 d) (OECD 301B; ISO 9439; 92/69/EEC, C.4-C) Readily biodegradable (according to OECD criteria).

Bioaccumulation potential

Assessment bioaccumulation potential:

Accumulation in organisms is not to be expected.

Other adverse effects

Adsorbable organically-bound halogen (AOX):

This product contains no organically-bound halogen.

Additional information

Add. remarks environm. fate & pathway:

Treatment in biological waste water treatment plants has to be performed according to local and administrative regulations.

Other ecotoxicological advice:

Inhibition of degradation activity in activated sludge is not to be anticipated during correct introduction of low concentrations.

13. Disposal Considerations

Must be disposed of or incinerated in accordance with local regulations.

Contaminated packaging:

Uncontaminated packaging can be re-used.

Packs that cannot be cleaned should be disposed of in the same manner as the contents.

14. Transport Information

Domestic transport:

Packing group: III
ID number: UN 3082
Transport hazard class(es): 9, EHSM
Proper shipping name: ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (contains FATTY ALCOHOL ALCOXYLATE)

Sea transport

IMDG

Packing group: III
ID number: UN 3082
Transport hazard class(es): 9, EHSM
Marine pollutant: YES
Proper shipping name: ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (contains FATTY ALCOHOL ALCOXYLATE)

BASF Safety data sheet
Date / Revised: 29.05.2021
Product: **Plurafac® LF 403**

Version: 5.0

(30044016/SDS_GEN_TH/EN)

Date of print 24.01.2024

Air transport

IATA/ICAO

Packing group:	III
ID number:	UN 3082
Transport hazard class(es):	9, EHSM
Proper shipping name:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (contains FATTY ALCOHOL ALCOXYLATE)

15. Regulatory Information**Other regulations**

16. Other Information

This product is of industrial quality and unless otherwise specified or agreed intended exclusively for industrial use. This includes the mentioned and recommended usage. Any other intended applications should be discussed with the manufacturer. In particular this concerns the application for products that are the object of special standards and regulations.

Vertical lines in the left hand margin indicate an amendment from the previous version.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.